

Junghee Kim

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Research Interests

- Resilient Transportation Systems
- Mobility Behavior Modeling
- Transportation Network Optimization
- Machine Learning for Transportation Systems
- Transportation Safety and Risk Analysis

Education

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Civil and Environmental Engineering, GPA: 3.45/4.3

- Thesis: Detecting Changes in People's Behavior Focusing on Seoul Subway Ridership During COVID-19 (Advisor: Yoonjin Yoon)

Daejeon, South Korea

Aug 2022 – Aug 2024

University of Seoul

B.S. in Transportation Engineering, GPA: 4.03/4.5

- Capstone Project: Introducing the Han River Flyer: A New Urban Transportation System (Advisor: Jae-Sung Choi)

Seoul, South Korea

Mar 2010 – Aug 2015

Research and Teaching Experience

Research Experience

Independent Research Experience: CNN-based Traffic Speed Prediction March 2025 – June 2025

- Developed a CNN-based traffic speed forecasting model by reformulating spatiotemporal traffic data into image-like representations
- Replicated and extended the methodology of Ma et al. (2017), adapting the framework to Seoul's arterial and urban road networks
- Conducted systematic sensitivity analyses across multiple forecasting horizons (10 and 20 min) and input window settings (30 and 40 min)
- Performed comparative evaluations between deep CNN architectures and a baseline MLP model, achieving up to 6.14% improvement in MSE (3.80% average gain)
- Found that a two-layer CNN achieved optimal performance, suggesting that simpler architectures better capture the relatively low-complexity spatiotemporal dynamics of traffic patterns in Seoul's urban core

Graduate Researcher, TRUE Lab(Transportation Research and Urban Engineering Laboratory),

KAIST(now Spacetime Intelligence Lab, StiL)

Daejeon, South Korea

Aug 2022 – Aug 2024

- Conducted data-driven research integrating Dynamic Time Warping and Hierarchical Clustering with large-scale transit data to analyze regional mobility patterns across Seoul
- Defined and categorized distinct phases of the COVID-19 pandemic based on changes in social distancing policies and variations in monthly average subway ridership decline rates to examine long-term impacts on urban mobility
- Analyzed mobility patterns from 232 subway stations in Seoul, identifying 6 distinct clusters; over 64% of stations belonged to clusters that had not returned to pre-pandemic (2019) daily ridership levels
- Proposed region-specific operational strategies and data-driven policy recommendations based on the identified mobility patterns, including targeted interventions for stations requiring mobility recovery, priority economic support, and enhanced public health management for stations with high essential travel demand

Teaching Experience

Teaching Assistant, Introduction to Data Science for Civil & Environmental Engineering, KAIST

Daejeon, South Korea

Sep 2023 – Dec 2023

- Reviewed lecture materials and course resources to support the instructor
- Evaluated and graded theoretical and Python-based assignments, providing detailed feedback on logical errors and deduction factors using Pandas and NumPy
- Led Q&A sessions for midterm and final exam preparation, assisting students with course-related inquiries

Professional Experience

YBS Co., Ltd.

Traffic Analytics / Machine Learning, Data Scientist

Seoul, South Korea

Sep 2025 – Nov 2025

- Built a two-stage XGBoost-based predictive framework for metro congestion during large-scale events, achieving up to 98% prediction accuracy by integrating meteorological data

- and demand features
- Identified critical tipping points in subway ridership sensitivity to meteorological conditions under event scenarios
- Translated model outputs into actionable insights through technical reports and presentations for key stakeholders
- Deployed the final system as a production-level web service for operational use by Seoul Metro

Korea Road Traffic Authority Seoul Branch

Seoul, South Korea

- Signal Operation Division (Seoul Metropolitan Police Agency), Assistant Manager Jan 2021 – July 2022
- Operated and managed Seoul's intersection signal control systems (SISO, UTMS) for the northern metropolitan region
 - Maintained the intersection database for Seoul and generated statistical reports on traffic signal operations
 - Responded to public inquiries regarding traffic signal timings and coordinated with police authorities to review and implement changes, including new intersection installations and signal phase adjustment

Korea Road Traffic Authority Daejeon, Sejong & Chungnam Branch

Daejeon, South Korea

- Safety Facilities Improvement Division, Assistant Manager May 2018 – Jan 2021
- Performed annual statistical analysis of traffic accident data to identify high-risk locations, targeting 22 intersections and 4 road segments per year
 - Conducted preliminary designs for the selected sites—including geometric modifications, installation of safety facility, and signal timing adjustments—achieving an average 56% reduction in traffic fatalities and a 33% reduction in overall accident rates
 - Assessed the feasibility of roundabout installations and evaluated and mitigated accident risks at irregular intersections
 - Reviewed and optimized signal operations to ensure operational appropriateness and safety
 - Executed road traffic safety audits for six major roads, including County Road No. 2 in Buyeo-gun

Safety Investigation and Inspection Division, Staff

Sep 2016 – April 2018

- Conducted on-site investigations to reconstruct traffic accidents and identify contributing factors
- Performed scientific cause analysis utilizing advanced simulation software (MADYMO, PC-Crash) and empirical field experiments
- Authored comprehensive traffic accident analysis reports by synthesizing field data and forensic evidence, including skid marks, CCTV footage, and vehicle collision traces

Publications & Presentation

Publications

- Kim, J.,** Lee, H., Kim, S., & Yoon, Y. (2025). "Time Series Cluster Analysis of Seoul Subway Usage Patterns under COVID-19 Based on Dynamic Time Warping." *Journal of the Korean Society of Transportation* (Peer-reviewed), Vol. 43, No. 1, pp. 15–37.
- Kim, J.** (2024). "Detecting Changes in People's Behavior: Focusing on Seoul Subway Ridership During COVID-19." (*Master's Thesis*, KAIST).

Presentations

- Kim, J.,** Kwak, E., Jung, H., & Choi, J. (2013). "Traffic Analysis: Transition of taxi into public transportation – Use of Bus Rapid Transit system of Taxi" Presented at the *68th Spring Conference of the Korean Society of Transportation (KST)*, Seoul, Korea; Awarded Best Paper (Student Category)

Awards & Honors

Awards

- Chairman's Award, Korea Road Traffic Authority** (Dec 2019, Dec 2020); Awarded for contributions to institutional goals based on individual annual performance
- Special Award (1st Place)**, Capstone Transportation Engineering Project, *The Seoul Institute* (May 2013); Awarded the top prize in the Department of Transportation Engineering at the 27th Urban Science Joint Exhibition; Recognized by the Seoul Institute for proposing an urban cable car system as an innovative transit alternative
- Best Paper Award (Student Category)**, *68th Spring Conference, Korean Society of Transportation* (Feb 2013); Title: "Traffic Analysis: Transition of taxi into public transportation – Use of Bus Rapid Transit system of Taxi"

Scholarships

- KAIST Support Scholarship (Graduate) & Research Assistantship, KAIST (2022 – 2024); Awarded a scholarship covering over 90% of tuition along with a monthly research stipend
- Municipal Mayor's Scholarship, University of Seoul (Fall 2013); Awarded to the top-ranked student in the department for outstanding academic performance
- Academic Excellence Scholarship (Type II), University of Seoul (Spring 2012)
- Academic Excellence Scholarship (Type II), University of Seoul (Fall 2011)

Skills

Language & Programming

- Python(Numpy, Pandas, Scikit-learn, Matplotlib, GeoPandas)
- MySQL
- QGIS
- AutoCAD

Certifications

- Certified Construction Technology Engineer (Intermediate Level), Korea Construction Engineers Association, 2021
- Traffic Accident Analyst Certification, Korea Road Traffic Authority, 2017
- Certified Traffic Engineer (Korean National Technical Qualification), Korea Industrial Human Resources Corporation, 2013

International Experience

Academic Visit

Nanyang Technological University

Singapore

Dec 2013

- Explored current research in Civil Engineering by attending faculty-led presentations and discussions
- Interacted with NTU students, fostering cross-cultural academic exchange

Exchange Student

University of North Carolina Pembroke

USA

Aug – Dec 2012

- Active member of Cru (Christian campus ministry organization), participating in weekly Bible study sessions
- Achieved GPA 3.93/4.0 during exchange semester

Other Activities

Member

SaRang Church

Seoul, South Korea

Sep 2024 – Present

- Participated in discussions and activities related to peace and unification on the Korean Peninsula
- Organized a Peace and Unification Program and provided piano accompaniment
- Participated in volunteer work at a rural church, contributing to sidewalk block installation

Member, Small Group Leader

Daeduk Hanbit Church

Daejeon, South Korea

Jul 2017 – Jan 2021, Aug 2022 – Aug 2024

- Participated in overseas field experience in Kazakhstan and in Korea-based programs related to North Korea, engaging in cross-cultural and regional learning activities
- Led weekly small-group discussions for a Bible study group, facilitating communication and peer engagement

Lab Manager

TRUE Lab(now StIL), KAIST

Daejeon, South Korea

Sep 2023 – July 2024

- Organized and managed weekly lab seminar schedules, including presenter assignments and session coordination
- Served as a primary liaison between the PI and lab members, streamlining internal communication and managing daily operations
- Implemented team-building initiatives, such as 'Cookie Time' and communal gardening, to foster a collaborative culture and strengthen interpersonal bonds
- Led lab tours for undergraduate students, introducing the lab's research history and ongoing projects
- Facilitated inter-laboratory networking by organizing exchange events to promote collaboration within the department

Dormitory Representative

Dasom Hall, KAIST

Daejeon, South Korea

Mar 2023 – Feb 2024

- Represented dormitory residents in official meetings with housing administration
- Identified student concerns through regular feedback collection and proposed improvement solutions
- Facilitated resolution of facility and welfare issues in coordination with dormitory management